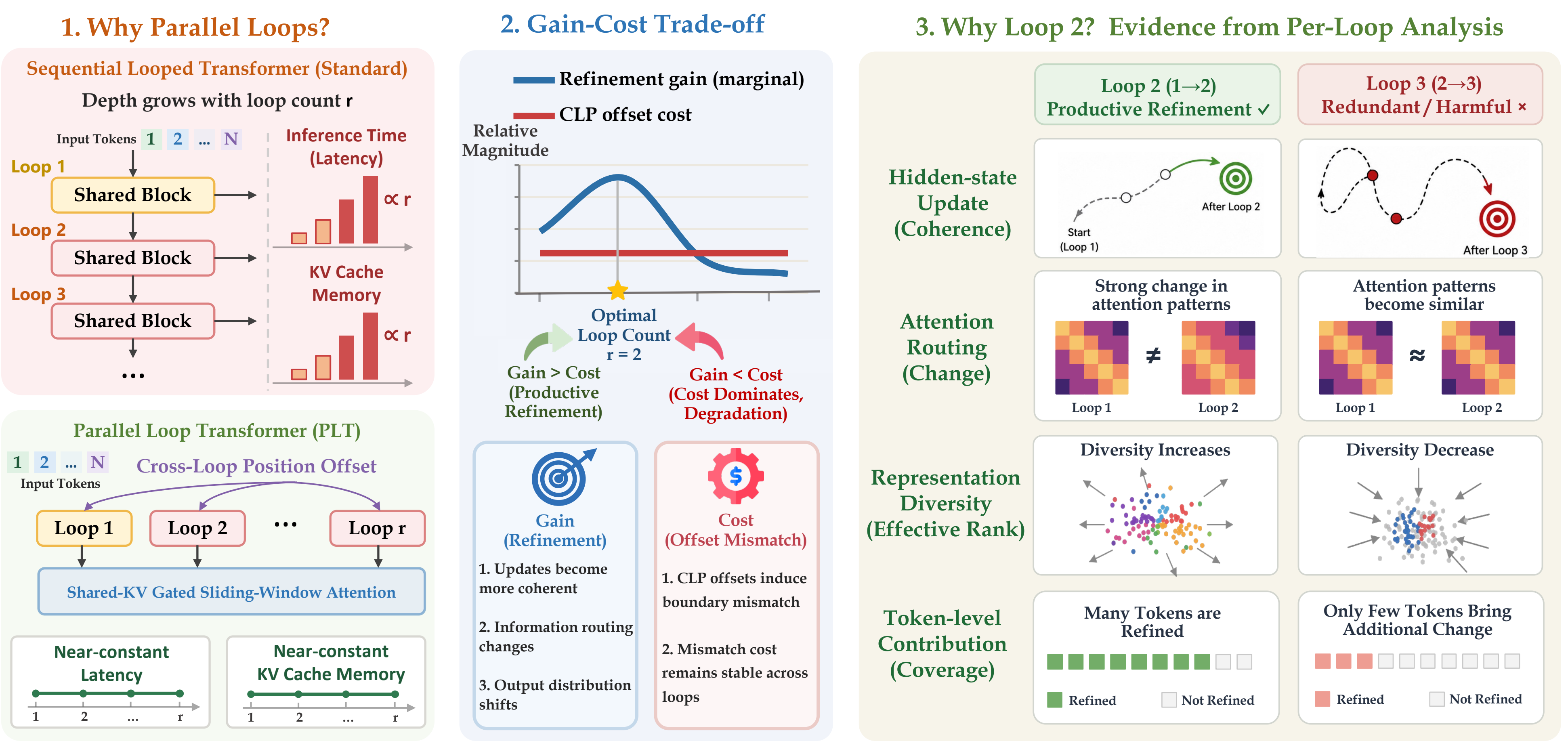

1. Why Parallel Loops?
Sequential Looped Transformer (Standard)
Depth grows with loop count r
Input Tokens 1 2 ... N
Loop 1
Shared Block
Loop 2
Shared Block
Loop 3
Shared Block
...
Inference Time (Latency)
∝ r
KV Cache Memory
∝ r
Parallel Loop Transformer (PLT)
1 2 ... N
Input Tokens
Cross-Loop Position Offset
Loop 1
Loop 2
...
Loop r
Shared-KV Gated Sliding-Window Attention
Near-constant Latency
1 2 ... r
Near-constant KV Cache Memory
1 2 ... r
2. Gain-Cost Trade-off
Refinement gain (marginal)
CLP offset cost
Relative Magnitude
Optimal Loop Count r = 2
Gain > Cost (Productive Refinement)
Gain < Cost (Cost Dominates, Degradation)
Gain (Refinement)
1. Updates become more coherent
2. Information routing changes
3. Output distribution shifts
Cost (Offset Mismatch)
1. CLP offsets induce boundary mismatch
2. Mismatch cost remains stable across loops
3. Why Loop 2? Evidence from Per-Loop Analysis
Loop 2 (1→2) Productive Refinement ✓
Loop 3 (2→3) Redundant / Harmful ×
Hidden-state Update (Coherence)
Start (Loop 1)
After Loop 2
After Loop 3
Attention Routing (Change)
Strong change in attention patterns
Loop 1 ≠ Loop 2
Attention patterns become similar
Loop 1 ≈ Loop 2
Representation Diversity (Effective Rank)
Diversity Increases
Diversity Decrease
Token-level Contribution (Coverage)
Many Tokens are Refined
Refined
Not Refined
Only Few Tokens Bring Additional Change
Refined
Not Refined